Applied Statistics (MATH10096)

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The `glm` function in `R`

- The `glm` function provides the means for using GLMs in `R`.
- Its use is similar to that of the `lm` function, with two differences.
- The right hand side of the model formula, specifying the form for the linear predictor, now gives the link function of the mean of the response, rather than the mean of the response directly.
- Also `glm` takes a `family` argument, which is used to specify the distribution from the exponential family to use, and the `link` function that is to go with it.

The `glm` function in R

Hen Harrier dataset: Gamma GLM

- Recall that we had

$$\mathbb{E}(c_i) \equiv \mu_i = \frac{\alpha d_i^3}{\delta + d_i^3}.$$

- Here c_i represents the consumption rate of Grouse by Hen Harriers (per day) and d_i is the corresponding Grouse density (per km^2).

- We previously observed that by using the inverse link function, the right hand-side of this equation can be expressed linearly in the parameters:

$$\frac{1}{\mu_i} = \frac{1}{\alpha} + \frac{\delta}{\alpha}\frac{1}{d_i^3} = \beta_1 + \beta_2 \frac{1}{d_i^3}, \quad \beta_1 = \frac{1}{\alpha}, \quad \beta_2 = \frac{\delta}{\alpha}$$

- Additionally, we argued that the Gamma distribution would be a plausible choice for modeling the consumption rate.

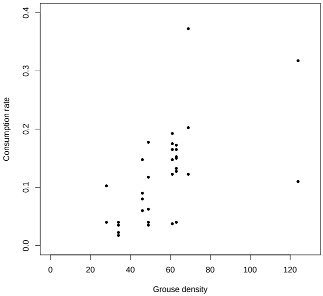
The glm function in R

Hen Harrier dataset: Gamma GLM

↪ Let us load and plot the data.

```
library(gamair)
data(harrier)
# Remove entries with zero consumption rate
harrier <- harrier[harrier$Consumption.Rate!=0, ]

plot(harrier$Grouse.Density, harrier$Consumption.Rate,
     ylim = c(0, 0.4), xlim = c(0, 130),
     xlab = "Grouse density", ylab = "Consumption rate", pch = 20)
```



The `glm` function in R

Hen Harrier dataset: Gamma GLM

↪ Now, fit the model.

```
hm <- glm(Consumption.Rate ~ I(1/Grouse.Density^3),
          family = Gamma(link = "inverse"), data = harrier)
summary(hm)
```

```
##
## Call:
## glm(formula = Consumption.Rate ~ I(1/Grouse.Density^3), family = Gamma(link = "inverse"),
##     data = harrier)
##
## Coefficients:
##                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)            4.676e+00  1.156e+00   4.046 0.000321 ***
## I(1/Grouse.Density^3) 5.386e+05  1.655e+05   3.255 0.002741 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Gamma family taken to be 0.347565)
##
##     Null deviance: 16.166  on 32  degrees of freedom
## Residual deviance: 10.472  on 31  degrees of freedom
## AIC: -92.383
##
## Number of Fisher Scoring iterations: 6
```

The `glm` function in `R`

Hen Harrier dataset: Gamma GLM

↪ We know that for the Gamma distribution, the dispersion parameter is not one and it must be estimated. One possible estimator is the Pearson estimator, given by the sum of the squared Pearson residuals divided by the number of degrees of freedom:

$$\hat{\phi} = \frac{1}{n-p}\sum_{i=1}^{n}\frac{(y_i - \hat{\mu}_i)^2}{V(\hat{\mu}_i)}.$$

↪ This is the estimator implemented by the `glm` function.

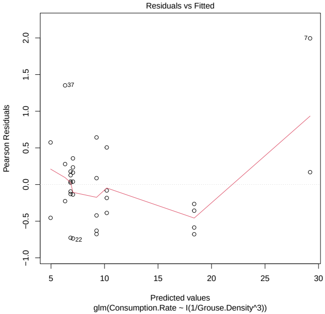
```
sum(residuals(hm, type = "pearson")^2)/(nrow(harrier) - 2)
## [1] 0.347565
summary(hm)$dispersion
## [1] 0.347565
```

The `glm` function in R

Hen Harrier dataset: Gamma GLM

↪ As with any model, we should check residuals before proceeding.

```
plot(hm, which = 1)
```

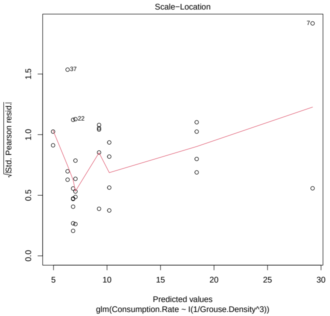


The `glm` function in `R`

Hen Harrier dataset: Gamma GLM

↪ As with any model, we should check residuals before proceeding.

```
plot(hm, which = 3)
```

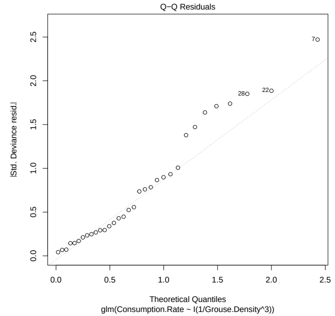


The `glm` function in `R`

Hen Harrier dataset: Gamma GLM

↪ As with any model, we should check residuals before proceeding.

```r
plot(hm, which = 2)
```

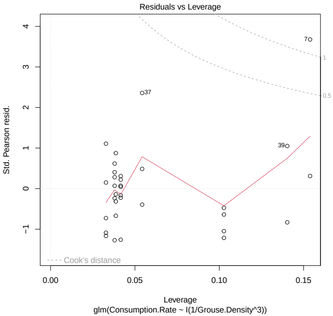


The `glm` function in R

Hen Harrier dataset: Gamma GLM

↪ As with any model, we should check residuals before proceeding.

```
plot(hm, which = 5)
```



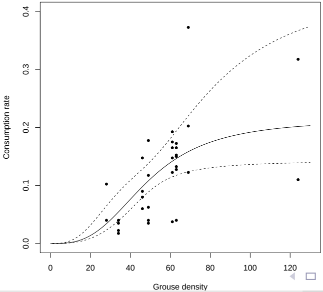
The `glm` function in R

Hen Harrier dataset: Gamma GLM

↪ For this simple example, it can help to plot model predictions and data on the same plot.

```
pd <- data.frame(Grouse.Density = 0:130) #data at which to predict
fv <- predict(hm, pd, type = "response") #get model predictions

lp <- predict(hm, pd, se = TRUE)
plot(harrier$Grouse.Density, harrier$Consumption.Rate,
     ylim = c(0, 0.4), xlim = c(0, 130),
     xlab = "Grouse density", ylab = "Consumption rate", pch = 20)
lines(0:130, fv)
lines(0:130, 1/(lp$fit + qt(0.975, df = (nrow(harrier) - 2)) *lp$se.fit), lty = 2)
lines(0:130, 1/(lp$fit - qt(0.975, df = (nrow(harrier) - 2)) *lp$se.fit), lty = 2)
```



The `glm` function in `R`

Hen Harrier dataset: Gamma GLM

- Is the cubic dependence on density the best model for these data? Let's try two models: one with linear dependence and one with quadratic dependence.

```
hm1 <- glm(Consumption.Rate ~ I(1/Grouse.Density),
          family = Gamma(link = "inverse"), data = harrier)

hm2 <- glm(Consumption.Rate ~ I(1/Grouse.Density^2),
          family = Gamma(link = "inverse"), data = harrier)
AIC(hm, hm1, hm2)
```

```
##      df       AIC
## hm    3 -92.38289
## hm1   3 -92.63887
## hm2   3 -94.17016
```

- The AIC slightly favours the model with a quadratic dependence on density.

- Note that to compare these three models we cannot use the generalized likelihood ratio test result because the models are not nested!

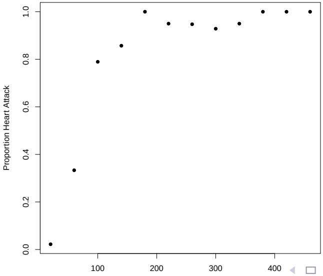
The glm function in R

Heart attacks example: Binomial GLM

- We revisit the heart attack data example discussed earlier.
- The goal is to construct a model to explain the proportion of patients suffering a heart attack based on the creatine kinase levels.

```
ck<- c(20 , 60 , 100 , 140 , 180 , 220 , 260 , 300 , 340 , 380 , 420 , 460)
ha <- c(2 , 13 , 30 , 30 , 21 , 19 , 18 , 13 , 19 , 15 , 7 , 8 )
ok <- c( 88 , 26 , 8 , 5 , 0 , 1 , 1 , 1 , 1  , 0 , 0  , 0)
heart <- data.frame(ck = ck, ha = ha, ok = ok)

p <- heart$ha/(heart$ha + heart$ok)
plot(heart$ck, p, xlab = "Creatinine kinase level", ylab = "Proportion Heart Attack",pch=16)
```



The `glm` function in `R`

Heart attacks example: Binomial GLM

$\hookrightarrow$ Recall that our basic model is

$$y_i \sim \text{Binomial}(N_i, p_i),$$

$\hookrightarrow$ Here y_i is the number of heart attack victims, observed out of N_i patients, with CK level x_i and p_i is the probability of heart attack at CK level x_i

$$p_i = \frac{e^{\beta_1+\beta_2 x_i}}{1+e^{\beta_1+\beta_2 x_i}}.$$

$\hookrightarrow$ We have that $\mathbb{E}(y_i) \equiv \mu_i = N_i p_i$ and

$$g(\mu_i) = \log\left(\frac{\mu_i}{N_i - \mu_i}\right) = \log\left(\frac{p_i}{1-p_i}\right) = \text{logit}(p_i) = \eta_i = \beta_1 + \beta_2 x_i.$$

The `glm` function in `R`

Heart attacks example: Binomial GLM

↪ In `R` there are two ways of specifying binomial models using the `glm` function.

```
# Approach 1
mod.0 <- glm(heart$ha/(heart$ha + heart$ok) ~ ck,
             family = binomial(link = logit),
             weights = heart$ha + heart$ok,
             data = heart)
mod.0

##
## Call:  glm(formula = heart$ha/(heart$ha + heart$ok) ~ ck, family = binomial(link = logit)
##     data = heart, weights = heart$ha + heart$ok)
##
## Coefficients:
## (Intercept)           ck
##    -2.75836      0.03124
##
## Degrees of Freedom: 11 Total (i.e. Null);  10 Residual
## Null Deviance:     271.7
## Residual Deviance: 36.93  AIC: 62.33
```

The `glm` function in `R`

Heart attacks example: Binomial GLM

↪ In `R` there are two ways of specifying binomial models using the `glm` function.

```
# Approach 2
mod.0 <- glm(cbind(ha,ok) ~ ck,
             family = binomial(link = logit),
             data = heart)
mod.0

##
## Call:  glm(formula = cbind(ha, ok) ~ ck, family = binomial(link = logit),
##     data = heart)
##
## Coefficients:
## (Intercept)           ck
##    -2.75836      0.03124
##
## Degrees of Freedom: 11 Total (i.e. Null);  10 Residual
## Null Deviance:      271.7
## Residual Deviance: 36.93   AIC: 62.33
```

The `glm` function in `R`

Heart attacks example: Binomial GLM

- As an example, the null deviance (from a model with only an intercept) and the residual deviance (from the fitted model) can be combined to give the proportion of deviance explained, a generalization of r^2, as follows:

```
1 - (summary(mod.0)$deviance/summary(mod.0)$null.deviance)
## [1] 0.8640893
```

- Notice that the deviance is quite high for the χ^2_{10} random variable that it should approximate if the model is fitting well.

- In fact,

```
pchisq(summary(mod.0)$deviance, df = (nrow(heart) - 2), lower.tail = FALSE)
## [1] 5.822516e-05
```

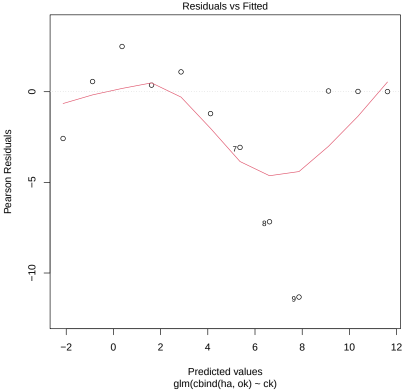
- There is a *very* small probability of a χ^2_{10} random variable being as large as 36.93.

The `glm` function in R

Heart attacks example: Binomial GLM

↪ Residual plots

```
plot(mod.0, which = 1)
```

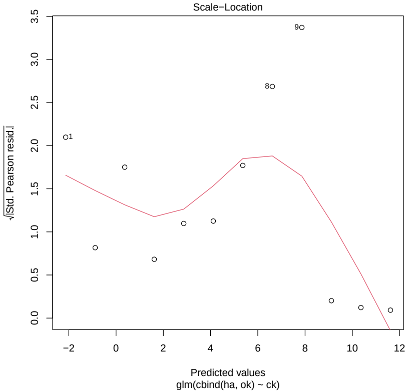


The `glm` function in R

Heart attacks example: Binomial GLM

↪ Residual plots

```
plot(mod.0, which = 3)
```

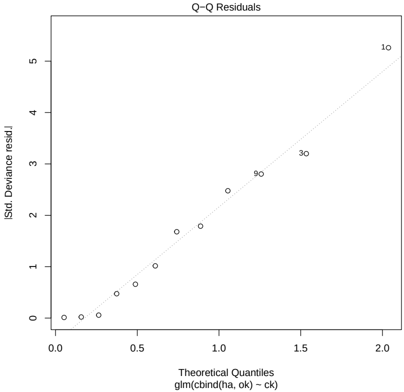


The `glm` function in R

Heart attacks example: Binomial GLM

$\hookrightarrow$ Residual plots

```
plot(mod.0, which = 2)
```

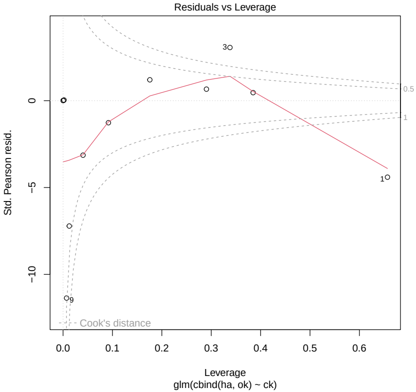


The `glm` function in R

Heart attacks example: Binomial GLM

↪ Residual plots

```r
plot(mod.0, which = 5)
```

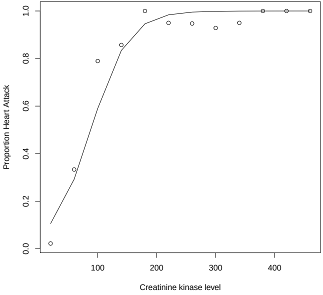


The `glm` function in R

Heart attacks example: Binomial GLM

↪ Notice how the problems do not stand out so clearly from a plot of the fitted values overlayed on the raw estimated probabilities.

```
plot(heart$ck, p,
     xlab = "Creatinine kinase level",
     ylab = "Proportion Heart Attack")
lines(heart$ck, fitted(mod.0))
```



↪ Note also that the fitted values provided by `glm` for binomial models are the estimated p_i's, rather than the estimated μ_i's $(= p_i N_i)$.

The `glm` function in R

Heart attacks example: Binomial GLM

- The trend in the mean of the residuals suggests trying a cubic model, rather than the initial straight line.

```
mod.1 <- glm(cbind(ha,ok) ~ ck + I(ck^2) + I(ck^3),
             family = binomial, data = heart)
mod.1
```

```
##
## Call:  glm(formula = cbind(ha, ok) ~ ck + I(ck^2) + I(ck^3), family = binomial,
##     data = heart)
##
## Coefficients:
## (Intercept)           ck      I(ck^2)      I(ck^3)
##  -5.786e+00    1.102e-01   -4.649e-04    6.448e-07
##
## Degrees of Freedom: 11 Total (i.e. Null);  8 Residual
## Null Deviance:      271.7
## Residual Deviance: 4.252   AIC: 33.66
```

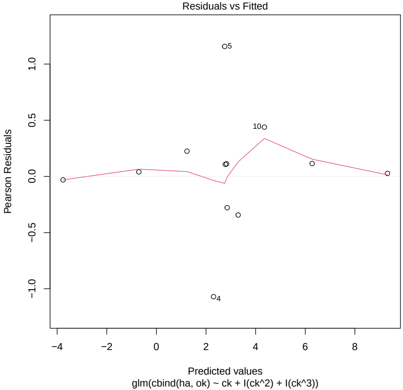
- Clearly, 4.252 is not too large for consistency with a χ^2_8 distribution (it is less than the expected value, in fact) and the AIC has improved (decreased) substantially.

The glm function in R

Heart attacks example: Binomial GLM

↪ Residual plots

```
plot(mod.1, which = 1)
```

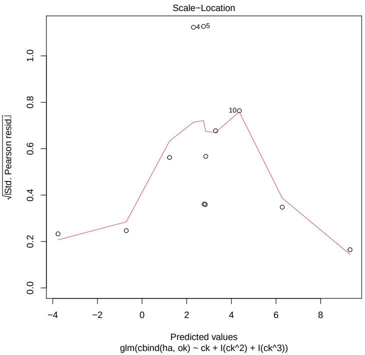


The `glm` function in `R`

Heart attacks example: Binomial GLM

↪ Residual plots

```
plot(mod.1, which = 3)
```



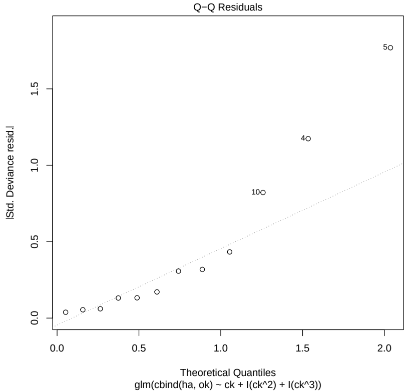
↪ If we had more data then such a departure from constant variance would be a cause for concern.

The `glm` function in R

Heart attacks example: Binomial GLM

↪ Residual plots

```
plot(mod.1, which = 2)
```

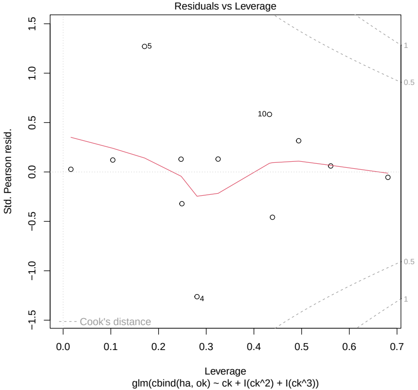


The `glm` function in R

Heart attacks example: Binomial GLM

↪ Residual plots

```
plot(mod.1, which = 5)
```

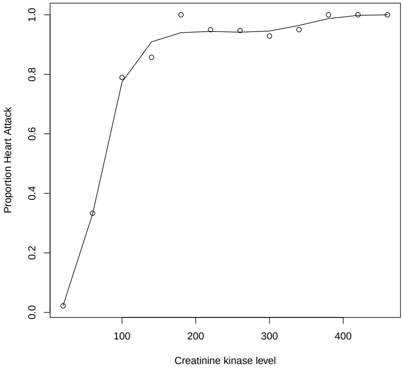


The `glm` function in R

Heart attacks example: Binomial GLM

↪ The fit is also clearly closer to the data now.

```
plot(heart$ck, p, xlab = "Creatinine kinase level",
     ylab = "Proportion Heart Attack")
lines(heart$ck, fitted(mod.1))
```



The `glm` function in R

Heart attacks example: Binomial GLM

↪ We can also use R to test the null hypothesis that `mod.0` is compatible with the data against the alternative that `mod.1` is required.

```
anova(mod.0, mod.1, test = "Chisq")

## Analysis of Deviance Table
##
## Model 1: cbind(ha, ok) ~ ck
## Model 2: cbind(ha, ok) ~ ck + I(ck^2) + I(ck^3)
##   Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
## 1        10     36.929
## 2         8      4.252  2   32.676 8.025e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

↪ A p-value this low indicates very strong evidence against the null hypothesis: we are rejecting that the coefficients associated with the quadratic and cubic effects are both zero.

The `glm` function in `R`

Heart attacks example: Binomial GLM

↪ Of course, we could still test whether the cubic effect is necessary or if a quadratic effect of CK suffices.

```
mod.2 <- glm(cbind(ha,ok) ~ ck + I(ck^2),
             family = binomial, data = heart)
anova(mod.2, mod.1, test = "Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: cbind(ha, ok) ~ ck + I(ck^2)
## Model 2: cbind(ha, ok) ~ ck + I(ck^2) + I(ck^3)
##   Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
## 1         9    15.4102
## 2         8     4.2525  1   11.158 0.0008368 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

↪ The cubic effect is indeed necessary.

The `glm` function in `R`

Logistic regression: melanoma case-control study

- Case-control studies are an important type of study, in which a group of patients with some disease (the cases) are compared to a randomly selected group of healthy subjects from the same population as the cases (the controls).
- Variables that might be associated with the disease are also collected for all subjects.
- If a variable is really associated with the disease then it ought to be predictive of whether a randomly selected patient in the study is a case or a control.
- Such predictivity can be assessed using GLMs of the 'logistic regression type'.

The `glm` function in `R`

Logistic regression: melanoma case-control study

- For example, consider a study looking at 143 cases of malignant melanoma (a serious skin cancer) in white male patients aged 25 to 55, classified according to skin type (A, B, or C for ‘celtic’, ‘middle european type’, or ‘Mediterranean’).
- Cases were compared to 356 white male controls aged 25-55 (selected without further reference to age or skin type).
- Patients were divided into 3 groups according to an ‘age’ factor variable, as well as being classified into 3 groups by the ‘skin’ factor variable
- There are therefore 9 groups in total.

The `glm` function in R

Logistic regression: melanoma case-control study

↪ The data are in a data frame called `md`.

```r
md <- data.frame(mel = c(15, 8, 7, 26, 18, 6, 30, 25, 8),
                 n = c(54, 52, 44, 75, 52, 42, 67, 66, 47),
                 age = rep(c("25-35", "35-45", "45-55"), each = 3),
                 skin = rep(c("A", "B", "C"), 3),
                 stringsAsFactors = TRUE)
md
```

```
##   mel  n   age skin
## 1  15 54 25-35    A
## 2   8 52 25-35    B
## 3   7 44 25-35    C
## 4  26 75 35-45    A
## 5  18 52 35-45    B
## 6   6 42 35-45    C
## 7  30 67 45-55    A
## 8  25 66 45-55    B
## 9   8 47 45-55    C
```

The `glm` function in R

Logistic regression: melanoma case-control study

- Consider testing the null hypothesis that skin type is not associated with melanoma, against the alternative that it is.

- Neglecting the possibility of an interaction term, we could do this by comparing the null model

$$\text{logit}(p_i) = \mu + \beta_j, \quad y_i \sim \text{binom}(p_i, n_i),$$

  to the alternative

$$\text{logit}(p_i) = \mu + \beta_j + \gamma_k, \quad y_i \sim \text{binom}(p_i, n_i),$$

- Here $j \in \{1, 2, 3\}$ corresponds to the age-band levels '25-35', '35-45', and '45-55', respectively, and $k \in \{1, 2, 3\}$ corresponds to skin types A, B, and C, respectively. Here, y_i is the number of melanoma cases out of n_i subjects (cases + controls) in group i, for $i = 1, \ldots, 9$.

The `glm` function in `R`

Logistic regression: melanoma case-control study

↪ Fitting these models and comparing them using a generalized likelihood ratio test (aka analysis of deviance) is straightforward.

```
mel0 <- glm(mel/n ~ age, family = binomial, data = md, weights = n)
mel1 <- glm(mel/n ~ age + skin, family = binomial, data = md, weights = n)
anova(mel0, mel1, test = "Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: mel/n ~ age
## Model 2: mel/n ~ age + skin
##   Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
## 1         6    20.5062
## 2         4     3.4389  2   17.067 0.0001967 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

↪ Here the p-value is very low: there is a very small probability of observing this large a difference in deviance between the two models if `mel0` really generated the data. This strongly suggests that `mel0` is not compatible with the data.

↪ In other words, there is strong evidence in favour of `mel1` and an effect of skin type on melanoma risk.

The `glm` function in `R`

Logistic regression: melanoma case-control study

↪ As an alternative to the `anova` function, we could have used the `drop1` function with the argument `test = "Chisq"`.

```
drop1(mel1, test = "Chisq")
```

```
## Single term deletions
##
## Model:
## mel/n ~ age + skin
##        Df Deviance    AIC     LRT  Pr(>Chi)
## <none>      3.4389 50.569
## age     2  12.3876 55.517  8.9487 0.0113976 *
## skin    2  20.5062 63.636 17.0673 0.0001967 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The `glm` function in `R`

Logistic regression: melanoma case-control study

- The next step is to examine the model coefficients to ascertain the nature and size of the effect. Let us use this example to illustrate how to interpret the coefficients in a logistic regression model.

- We first note that we need to impose identifiability constraints (as is always the case when factors are involved). We know that by default `R` sets the first level of each factor variable to the baseline, which corresponds to removing the associated columns from the design matrix.

- The first levels are skin type A and age band 25-35.

```
levels(md$age)
## [1] "25-35" "35-45" "45-55"
levels(md$skin)
## [1] "A" "B" "C"
```

The `glm` function in R

Logistic regression: melanoma case-control study

↪ Alternatively, we could check the model matrix.

```
model.matrix(mel1)

##   (Intercept) age35-45 age45-55 skinB skinC
## 1           1        0        0     0     0
## 2           1        0        0     1     0
## 3           1        0        0     0     1
## 4           1        1        0     0     0
## 5           1        1        0     1     0
## 6           1        1        0     0     1
## 7           1        0        1     0     0
## 8           1        0        1     1     0
## 9           1        0        1     0     1
## attr(,"assign")
## [1] 0 1 1 2 2
## attr(,"contrasts")
## attr(,"contrasts")$age
## [1] "contr.treatment"
##
## attr(,"contrasts")$skin
## [1] "contr.treatment"
```

The `glm` function in `R`

Logistic regression: melanoma case-control study

$\hookrightarrow$ Let us write the model containing both the age and skin type effects using dummy variables:

$$\text{logit}(p_i) = \mu + \beta_1 \text{age}_{35-45,i} + \beta_2 \text{age}_{45-55,i} + \gamma_1 \text{skin}_{B,i} + \gamma_2 \text{skin}_{C,i}, \quad i = 1, \ldots, 9,$$

where

$$\text{age}_{35-45,i} = \begin{cases} 1, & \text{if group i is in age-band level 35-45,} \\ 0, & \text{otherwise,} \end{cases}$$

$$\text{age}_{45-55,i} = \begin{cases} 1, & \text{if group i is in age-band level 45-55,} \\ 0, & \text{otherwise,} \end{cases}$$

$$\text{skin}_{B,i} = \begin{cases} 1, & \text{if skin type is B for group i,} \\ 0, & \text{otherwise,} \end{cases}$$

$$\text{skin}_{C,i} = \begin{cases} 1, & \text{if skin type is C for group i,} \\ 0, & \text{otherwise.} \end{cases}$$

The `glm` function in `R`

Logistic regression: melanoma case-control study

- What is the interpretation of β_1? It represents the difference in log odds of melanoma between the group whose age-band level is 35-45 and the reference age group (age band level 25-35), with both groups sharing the same skin type.

- To see why, and for the sake of concreteness, assume that the skin type for both groups is B. We are then effectively comparing groups 2 and 5 in the dataframe.

$$\text{logit}(p_5) = \log\left(\frac{p_5}{1-p_5}\right) = \mu + \beta_1 \times 1 + \beta_2 \times 0 + \gamma_1 \times 1 + \gamma_2 \times 0 = \mu + \beta_1 + \gamma_1,$$

$$\text{logit}(p_2) = \log\left(\frac{p_2}{1-p_2}\right) = \mu + \beta_1 \times 0 + \beta_2 \times 0 + \gamma_1 \times 1 + \gamma_2 \times 0 = \mu + \gamma_1.$$

- Therefore,

$$\log\left(\frac{p_5}{1-p_5}\right) - \log\left(\frac{p_2}{1-p_2}\right) = (\mu + \beta_1 + \gamma_1) - (\mu + \gamma_1) = \beta_1.$$

- It is easy to see that if replace skin type B with A or C, we would still obtain β_1.

The `glm` function in `R`

Logistic regression: melanoma case-control study

- We further have

$$\log\left(\frac{p_5}{1-p_5}\right) - \log\left(\frac{p_2}{1-p_2}\right) = \beta_1 \Rightarrow \log\left(\frac{\frac{p_5}{1-p_5}}{\frac{p_2}{1-p_2}}\right) = \beta_1 \Rightarrow \frac{p_5}{1-p_5} = \frac{p_2}{1-p_2}e^{\beta_1}.$$

- That is, for two groups with the same skin type, the odds of melanoma for individuals in age band 35-45 is e^{β_1} times the odds for those in the reference age group (25-35).

- Similarly, for two groups with the same skin type, the odds of melanoma for individuals in the age band 45-55 is e^{β_2} times the odds for those in the reference age group (25-35).

- For two groups in the same age-band, the odds of melanoma for individuals with skin type B is e^{γ_1} times the odds for those in the reference skin type group (i.e., skin type A).

- Finally, for two groups in the same age-band, the odds of melanoma for individuals with skin type C is e^{γ_2} times the odds for those in the reference skin type group (skin type A).

The `glm` function in R

Logistic regression: melanoma case-control study

↪ Let's look at the coefficients.

```
mel1

##
## Call:  glm(formula = mel/n ~ age + skin, family = binomial, data = md,
##     weights = n)
##
## Coefficients:
## (Intercept)     age35-45     age45-55        skinB        skinC
##     -1.0228       0.4766       0.7655      -0.2881      -1.1063
##
## Degrees of Freedom: 8 Total (i.e. Null);  4 Residual
## Null Deviance:      29.89
## Residual Deviance: 3.439   AIC: 50.57

exp(coefficients(mel1))

## (Intercept)    age35-45    age45-55       skinB       skinC
##   0.3595763   1.6106085   2.1499647   0.7496932   0.3307686
```

The `glm` function in `R`

Logistic regression: melanoma case-control study

- We conclude that for two groups with the same skin type, the estimated odds of melanoma for individuals in age band 35-45 are 1.61 times the odds for those in the reference age group (25-35). Therefore, odds of disease increase when the age band moves from baseline to the 'next level'.

- The estimated increase in odds is even greater when comparing the baseline age band to the 45-55 age band, as $e^{\beta_2} = 2.15$, assuming, of course, that skin type remains the same for both groups.

- On the other hand, for two groups in the same age-band, the estimated odds of melanoma for individuals with skin type B are 0.75 times the odds for those with skin type A, indicating a decrease in the odds of disease.

- Finally, for two groups in the same age-band, the estimated odds of melanoma for individuals with skin type C are 0.33 times the odds for those with skin type A, representing an even more pronounced decrease in the odds of melanoma.

The `glm` function in `R`

Logistic regression: melanoma case-control study

- We should attach confidence intervals to these statements and we are already in position to do so.
- We know from previous lectures that the maximum likelihood estimators of the parameters follow, in the large sample limit, a (multivariate) normal distribution with a mean vector equal to the true (but unknown) parameters and with a covariance matrix given by $\phi(\mathbf{X}^{\mathsf{T}}\mathbf{W}\mathbf{X})^{-1}$.
- Because for the binomial distribution, $\phi = 1$, this result can be used directly to find confidence intervals for the parameters.
- For instance, we know that a $100(1-\alpha)\%$ confidence interval for β_1 is given by

$$(\hat{\beta}_1 - z_{1-\alpha/2}\sigma_{\hat{\beta}_1}, \hat{\beta}_1 + z_{1-\alpha/2}\sigma_{\hat{\beta}_1}),$$

  where $\sigma_{\hat{\beta}_1}$ is one of the diagonal entries of the covariance matrix $(\mathbf{X}^{\mathsf{T}}\mathbf{W}\mathbf{X})^{-1}$.
- We then trivially have that a $100(1-\alpha)\%$ confidence interval for e^{β_1} is

$$(e^{\hat{\beta}_1 - z_{1-\alpha/2}\sigma_{\hat{\beta}_1}}, e^{\hat{\beta}_1 + z_{1-\alpha/2}\sigma_{\hat{\beta}_1}}).$$

The `glm` function in `R`

Logistic regression: melanoma case-control study

```
summary(mel1)

##
## Call:
## glm(formula = mel/n ~ age + skin, family = binomial, data = md,
##     weights = n)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -1.0228     0.2355  -4.343 1.41e-05 ***
## age35-45      0.4766     0.2689   1.773  0.07630 .
## age45-55      0.7655     0.2611   2.932  0.00337 **
## skinB        -0.2881     0.2267  -1.271  0.20381
## skinC        -1.1063     0.2829  -3.911 9.19e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 29.8893  on 8  degrees of freedom
## Residual deviance:  3.4389  on 4  degrees of freedom
## AIC: 50.569
##
## Number of Fisher Scoring iterations: 4

lower_bound <- exp(0.4766 - qnorm(0.975)*0.2689); upper_bound <- exp(0.4766 + qnorm(0.975)*0.2689)
lower_bound; upper_bound
```

The `glm` function in `R`

Logistic regression: melanoma case-control study

↪ Note that the confidence interval contains the value $1 = e^0$.

↪ Confidence intervals for the other parameters are obtained in a similar manner

↪ In fact, we can compute all intervals automatically using the `confint.default` function.

```
exp(confint.default(mel1))

##                 2.5 %    97.5 %
## (Intercept) 0.2266304 0.5705107
## age35-45    0.9508683 2.7280957
## age45-55    1.2887860 3.5865911
## skinB       0.4807427 1.1691073
## skinC       0.1899965 0.5758417
```

The `glm` function in R

Logistic regression: melanoma case-control study

- Case-control studies allow for estimation of all regression coefficients with the exception of the intercept, which in this case corresponds to the log odds of melanoma for group 1, the baseline group (i.e., individuals in the 25-35 age band with skin type A).
- This is because we have chosen the ratio of cases to controls rather than observing it directly in the population of interest.
- Therefore μ does not provide an estimate of the population log odds of disease (melanoma) for the baseline group.
- As a consequence, we cannot estimated the probabilities of disease for each group as they involve μ.

$$p_i = \frac{e^{\mu+\beta_1 \text{age}_{35-45,i}+\beta_2 \text{age}_{45-55,i}+\gamma_1 \text{skin}_{B,i}+\gamma_2 \text{skin}_{C,i}}}{1+e^{\mu+\beta_1 \text{age}_{35-45,i}+\beta_2 \text{age}_{45-55,i}+\gamma_1 \text{skin}_{B,i}+\gamma_2 \text{skin}_{C,i}}}, \quad i = 1, \ldots, 9.$$

The `glm` function in `R`

Logistic regression: melanoma case-control study

- ↪ The other regression coefficients, being log odds ratios, can be estimated regardless of the study design.

- ↪ Furthermore, estimates of the standard deviations of estimated slope coefficients and hypothesis testing techniques, such as the generalized likelihood ratio test, can still be used with case-control data.